

Research paper

Numerical investigation of balling defects in laser-based powder bed fusion of metals with Inconel 718

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ABSTRACT

Balling is one of the main single melt track defects limiting stable production of dense parts with laser-based powder bed fusion of metals (PBF-LB/M) additive manufacturing. A fundamental understanding of balling therefore is critical for process control and optimization. Violent interface dynamics present a severe challenge for mesh-based numerical schemes, and can be dealt with by particle methods. In this study, we focus on investigating single melt tracks by smoothed particle hydrodynamics (SPH). The considered regime is characterized by long liquid melt pools and remelting of the previous layer. The balling mechanism is driven by Marangoni and capillary forces. The resulting convective melt flux leads to the accumulation of melt in the tail region and leaves an exposed area of marginal layer build-up. In addition to the competing mechanisms of solidification and spreading, a competition between spreading and melt contraction shortly after the melt pinch-off is observed. We find a similar balling behavior yet different ball size at different laser power settings. We stress that SPH with accurate interface modeling has the capability to quantify balling processes and to improve the understanding of balling mechanisms.

1. Introduction

Laser-based powder bed fusion is becoming an increasingly competitive manufacturing technique. A high energy laser beam is applied to selectively melt powder particles within a powder bed. The molten particles coalesce to form a melt pool that subsequently fuses and consolidates with the previous layer of material. With an appropriate selection of process parameters a successive layer-by-layer buildup allows for the fabrication of dense parts with high design flexibility, for instance, by enabling complex part production down to the micro scale and tailored material properties. Powder bed fusion is applied to a variety of materials such as alloys [1,2], ceramics [3] and polymers [4] and is also utilized in the development of new material composites [5,6]. Hence, laser-based powder bed fusion is a promising addition to established formative and subtractive manufacturing approaches.

The fundamental building unit in laser-based powder bed fusion is the single melt track since each manufactured layer is composed of consecutive single tracks. Manufacturing defects of final parts often originate from melt-flow instabilities during single track formation. It is important to understand the defect generation in single track melt pools and their dependence on process parameters. Much effort has been dedicated to identify process maps that lead to stable single melt

tracks [7–9], where a known process map constraint is due to melt pool balling.

Balling describes the breakup of continuous melt pools into melt agglomerates. The process is also referred to as 'aggregation of melt islands' or 'humps' [10,11]. It is found to occur in melt tracks of quite different materials such as aluminum [12], titanium [9], nickel [13], and steel [14] alloys. As balling is observed at high laser scan speeds in combination with both high [15] and low [10] laser power, as well as low laser scan speeds and high laser power [16], it is considered as one of the key defect mechanisms [1]. Large-size balling can cause a non-uniform powder distribution in the consecutive powder layer, poor wettability of the previous layer [2,3], pore formation [14], and even an interruption of the build process due to a collision with the paving roller [14]. Final parts show a rough surface morphology [14], higher porosity [15] or poor inter layer bonding [17] around balls.

Several factors are identified to affect balling in single track melt pools. In general, balling occurs due to a minimization of the surface free energy by capillary forces. The overall intensity of capillary forces depends on melt temperature as, for many materials, surface tension decreases with increasing temperature. Consequently, Marangoni

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flow induced by high temperature gradients at the melt pool surface contributes to the breakup of continuous tracks [18]. Balling is further driven by poor wettability between the liquid metal and its surrounding powder particles and substrate [12,16,17,19]. Insufficient remelting of the previous layer is particularly problematic when oxide layers are formed on the substrate, as they increase the wetting angles between the liquid and solid material [20]. Sufficient remelting therefore is important to destroy such oxide layers [21] as well. Additionally, oxidation can invert surface tension gradients leading to surface flows directed towards the melt pool center and thus support melt aggregation [6]. At low power density settings balling can occur in the form of melt island-aggregates. Insufficient heating leads to partially melted particles, unmelted substrate beneath the powder, and a melt pool that tends to cluster rather than wet the surface [22,23]. Low melt temperatures and thus high melt viscosity and surface tension further enhance the process [1,5,14]. A commonly suggested solution to avoid such events is to increase the energy input [1] or reduce the powder layer thickness [9]. At high scan velocities, the melt pool width and thus the wetted substrate area decreases [24]. Long and narrow melt pools are prone to the Plateau-Rayleigh instability [2,3,7,10,13,15–18,24–26]. Poor melt pool spreading and shallow melt pool penetration further enhance the pool instability and breakup [7,12,14]. Analytical stability models [7,11,21,27] are used as breakup conditions for continuous single melt tracks [12,28–30], yet their ability to predict the onset of balling is still in question [31]. As the melt-pool-length-to-diameter ratio increases with increasing laser scan velocity, the melt pool is particularly susceptible to Plateau-Rayleigh instabilities in the process window of high scan speed and high laser power [24]. Ball formation also is found at low scan speeds and high energy input. High energy content within the melt pool reduces the rate of solidification. Hence, the time scale at which surface tension affects pool breakup increases as well [30]. This type of balling at large liquid formation is referred to as 'self balling' [16,32]. Ball formation itself is identified as a competitive non-equilibrium process between melt spreading and solidification [8,33,34]. Final droplet shapes depend on the complex interplay of capillary forces, inertia, solidification, thermal properties and heat content [33]. While balling is avoided when the melt can spread out before it solidifies, quick cooling rates lead to premature termination of spreading and enhance balling [8].

Extensive research is required to better understand the multitude of simultaneous mechanisms and to identify and enhance process maps for stable production. Process complexity in combination with temporal and spatial scales at the micro level make experimental in-situ measurements infeasible. Hence, increasing attention is devoted to numerical investigations [3,9,16,17,22,25,35–37]. An adequate representation of selective laser melting at the powder scale, however, requires comprehensive numerical modeling. Its multi-phase nature in combination with phase transitions and complex interface topologies with strong deformations pose severe challenges to the underlying algorithms.

Yet, several numerical studies are able to reproduce the balling phenomenon and allow for an in-depth analysis of the melt-pool breakup process. Ren et al. [22] apply the finite volume method in combination with a volume-of-fluid interface algorithm and ray-tracing laser model to study the absorptive behavior and flow kinetics of a copper alloy. Their single melt track study shows intense absorptivity oscillations, which are ascribed to balling and result in increased exposure of substrate rather than melt to the laser beam. A discrete element method including viscous coalescence and densification models has been used for the study of balling phenomena of a nickel alloy by Boutaous et al. [35]. Large variations of melt track widths indicate the presence of balling. From a sensitivity study on surface tension and viscosity they conclude that neither of the two have a significant influence on balling as coalescence of the melt is too fast. Li et al. [17] apply a phase-field method to 2D simulations of single melt tracks for pure nickel powder on stainless steel substrate. Their study of various wetting conditions between melt and substrate reveals a decreasing frequency of melt

pool breakup for low wetting angles. Single melt track simulations of 316L stainless steel show discontinuous tracks and melt agglomeration at partially melted powder particles [36]. The finite volume method algorithm by Gu et al. [37] is coupled with the volume of fluid interface model to study melt pool behavior in multi-layer scenarios. They show that the melt track of the first layer is discontinuous and exhibits melt agglomerates. A subsequent second layer leads to further clustering of melt and lack of fusion between the two layers. While at an increased power density a wavy but continuous first melt track is formed, in the second layer balling still occurs. It is concluded that the surface morphology of the first layer contributes to balling of consecutive layers. Single melt tracks of AlSi10Mg are investigated by Yu et al. [16] with a finite volume algorithm and volume of fluid interface treatment. They find melt pool breakup at high power densities. A detailed analysis of heating, melting and coalescing process of two powder particles reveals a flow pattern dominated by buoyancy and capillary forces. In [3], Moniz et al. apply a finite element method and the level-set interface method to study laser beam melting of Al₂O₃ oxide ceramics. They find the frequency of balling to increase with a reduction of power densities. They point out the lack of layer build up between balls due to the melt agglomeration. Balling of single melt tracks of titanium alloy Ti6Al4V is investigated by Yan et al. [9]. The results show partial melting of powder particles and a lack of substrate melting. The study of different powder setups reveals an independence of balling from the random powder packing, yet a strong dependence on the powder size distribution and powder layer thickness.

Most numerical research on balling processes is focused on balling in a regime characterized by low power densities, partially melted powder particles and small melt pools. Results show only slight or no melting of the substrate [10,16,17,22]. In this regime the low amount of melt commonly is given as reason for melt pool balling [1,3,16,36,37]. However, balling also occurs when there is full powder- and considerable substrate-melting [38]. Yet, a detailed numerical study of balling in this important part of the process map still is missing from literature. A material of particular interest for aerospace applications is the Inconel 718 nickel-based superalloy. Like many other alloys, the process map suffers from limitations in process parameter choice due to balling. Both, analytically [28] and experimentally [38] estimated power-scan-speed process maps including a balling threshold are available in literature. Only few numerical studies on balling can be found, none of which focuses on the analysis of driving forces of balling. Additionally, the majority of numerical research into balling is conducted with mesh based methods such as finite volume or finite element methods.

The aim of this work is to close the gap of balling research in the regime defined by a fully melted powder bed and considerable substrate remelting. We focus on the contribution of driving forces on the breakup mechanism and the ball formation in this regime. Thus, the research into balling of Inconel 718 melt tracks is extended by a more quantitative comparison of forces. To that end, we conduct several single melt track simulations at different laser power and laser scan speed settings. First, the simulation results are validated by comparison of melt pool depth and width with experimental data. Then, we focus on the balling process and compare simulation results at process parameter sets both with the presence and absence of balling to analytical and experimental references followed by an analysis of the underlying physical mechanisms. We also demonstrate the capability of smoothed particle hydrodynamics (SPH) to resolve such complex multi-phase phenomena, which to the best of our knowledge, has not been previously applied to the problem of balling in laser powder bed fusion. The remainder of this work is as follows: In section two, the physical model is introduced, relevant simplifications are stated and the numerical setup is described. Simulation results are shown and discussed in section three. Concluding remarks and observations are presented in the closing section.

2. Materials and methods

In this work smoothed particle hydrodynamics (SPH) is applied [39]. In contrast to finite volume or finite element methods it implicitly tracks the interface without requiring computationally demanding interface location and tracking algorithms making it particularly suitable for simulations that involve complex interfaces movements. SPH particles convect with the flow field eliminating the need for complex re-meshing or moving meshes. The method also is highly adaptable for simulating complex powder bed structures and fluid–solid boundaries, as different particles can be used for different materials. Additionally, the SPH method is stable for strong interface deformations, which makes it an ideal choice for simulating dynamic melt-pools and the balling process. For the accurate prediction of melt pool morphology and balling a WCSPH multi-phase model [40] is employed, extended by transport-velocity [41] and consistent boundary conditions [42]. For an accurate representation of melt pool dynamics, pressure, viscous, and gravity forces are taken into account. Evaporation effects on the melt are considered by a recoil-pressure model [43]. To account for the driving forces of balling, both capillary and Marangoni forces as well as wetting conditions are incorporated at the melt surface following [44]. A wetting angle of 70 degrees is chosen based on experimental contact angles measurements [8,45] and the cross section morphology in [38]. A precise representation of the energy balance is achieved by modeling thermal conduction in melt, powder and substrate [46], convective cooling by an ambient gas, evaporative cooling, a laser heat source, and solid–liquid phase change, considering the latent heat of melting and solidification. Material models from literature are applied to account for the temperature dependence of viscosity, surface tension, thermal conductivity, heat capacity and laser absorptivity, see A.3.

No ambient gas flow is induced and the surrounding gas is treated as isothermal to represent convective cooling. Thermal expansion within the melt pool and possibly resulting buoyancy forces are ignored. Thermal radiation and viscous heating are discarded. The applied absorptivity model accounts for the incidence angle, the dependence on temperature and an energy input at the material surface. As it does not consider multi-reflections within the vapor cavity, absorptivity is increased to match the target melt pool dimensions. While recoil pressure and evaporative cooling are included, mass loss due to evaporation is neglected. To account for increased viscosity of the melt in the mushy zone and to stabilize the simulations the dynamic viscosity is increased by a factor of ten, following [47,48]. The powder layer is explicitly resolved, yet powder particle motion is not incorporated at this stage.

The conservation laws of mass, momentum and energy govern the fluid mechanics and energy balance of balling. For incompressible fluids, the set of equations in Lagrangian form is

$$\frac{d\rho}{dt} = -\rho \nabla \cdot \mathbf{v}, \quad (1)$$

$$\frac{d\mathbf{v}}{dt} = \mathbf{F}^p + \mathbf{F}^v + \mathbf{F}^\sigma + \mathbf{F}^{ev} + \mathbf{F}^g, \quad (2)$$

$$\frac{dh}{dt} = Q^\lambda + Q^{ev} + Q^l, \quad (3)$$

where ρ , $\mathbf{v}$ and h are density, velocity, and enthalpy, respectively. Lagrangian-particle acceleration is due to pressure ($\mathbf{F}^p$), viscous ($\mathbf{F}^v$), surface tension ($\mathbf{F}^\sigma$), recoil pressure ($\mathbf{F}^{ev}$) and gravity ($\mathbf{F}^g$) forces. Changes in enthalpy are induced by thermal conduction (Q^λ), evaporative cooling (Q^{ev}) and laser irradiation (Q^l).

Pressure forces follow from gradients of the pressure field p as

$$\mathbf{F}^p = -\frac{1}{\rho} \nabla p. \quad (4)$$

Viscous forces are determined by the divergence of the strain rate tensor as

$$\mathbf{F}^v = \frac{1}{\rho} \nabla \cdot (\mu \nabla \mathbf{v}), \quad (5)$$

where μ is the temperature dependent dynamic viscosity. At the melt ambient-gas interface, both capillary and Marangoni forces,

$$\mathbf{F}^\sigma = \frac{1}{\rho} [\sigma \kappa \mathbf{n} + \nabla_s \sigma], \quad (6)$$

are applied. Here, σ , κ , $\mathbf{n}$, and ∇_s are the surface tension coefficient, interface curvature, interface normal vector and surface tangential gradient operator, respectively. Note that, no special interface capturing and tracking algorithms are needed as SPH naturally handles different phases with separate particles. Wetting of melt on the substrate and powder is modeled by Young's equation

$$\sigma^{solid-gas} = \sigma^{solid-liquid} + \sigma^{liquid-gas} \cos \theta, \quad (7)$$

where an equilibrium between solid, liquid and gas surface tension is achieved at the wetting angle θ . To account for the recoil pressure force we apply an approach based on the Clausius-Clapeyron equation for ideal gases assuming a constant latent heat of phase change

$$\mathbf{F}^{ev} = \frac{1}{\rho} A c_1 p_\infty \exp\left(\frac{\Delta H_{mol}}{R} \left(\frac{1}{T} - \frac{1}{T_{ev}}\right)\right), \quad (8)$$

with the evaporating surface area A , coefficient $c_1 = 0.54$, ambient pressure p_∞ , molar latent heat of evaporation ΔH_{mol} , universal gas constant R , temperature T and boiling temperature T_{ev} , respectively [49]. Finally, gravity g acts as body force

$$\mathbf{F}^g = -g. \quad (9)$$

On the right-hand-side of the energy equation, Fourier's law of thermal conduction

$$Q^\lambda = \nabla \cdot (\lambda \nabla T), \quad (10)$$

with thermal conductivity λ is used in the melt, gas, and solid phase. Evaporative cooling of the melt is determined from the enthalpy flux of the evaporating material. Here, the mass flux approximation from [49] is given by

$$J_{ev} = c_2 c_3 p_r \sqrt{\frac{M}{2\pi R T}}, \quad (11)$$

where the coefficient $c_2 = 0.82$ and the sticking coefficient $c_3 = 1$ are chosen according to [49]. In Eq. (11), p_r is the recoil pressure and M the atomic mass of the evaporating (melt-) material. Evaporative cooling follows as

$$Q^{ev} = -A (h + h_{ev}) J_{ev}, \quad (12)$$

where both the enthalpy h and the specific heat of evaporation h_{ev} are considered. This heat sink is applied at the interface between melt and the ambient gas phase. Finally, laser irradiation is modeled by a Gaussian energy source term

$$Q^l = -\alpha \Omega A \frac{8P}{\pi d_s^2} \exp\left(-8 \left(\frac{r_0 - r}{d_s}\right)^2\right) \quad (13)$$

with absorptivity α , directional absorptance correction Ω , laser power P , laser spot diameter d_s , laser axis location r_0 and radial coordinate r . Ω is approximated by Schlick's approach $\Omega = 1 - (1 - \cos(\phi))^5$, where ϕ is the angle between the laser axis and the incident area normal vector [50]. The energy deposition is considered to occur at the phase interface only [43].

Phase change is modeled in SPH by simply transforming particles from one phase to another. As criterion, we use an enthalpy threshold where latent heat of fusion and solidification is considered within the threshold. The rate of temperature change in the mushy zone between the two states is considered through specific heat capacity models, see A.3.

The set of equations is closed by Cole's [51] equation of state

$$p = p_0 \left(\left(\frac{\rho}{\rho_0} \right)^\gamma - 1 \right). \quad (14)$$

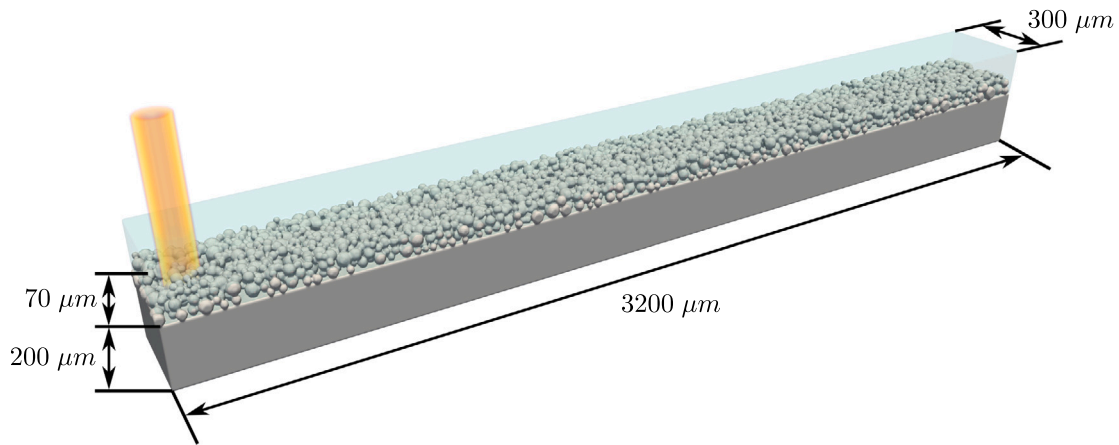


Fig. 1. Domain dimensions of substrate, powder and ambient gas. Laser scan path from $x = 0$ to $x = 3000 \mu\text{m}$.

Table 1
Single melt track simulation parameters.

Serial number	Laser power [W]	Scan speed [mm/s]	LED [J/m]	VED [J/mm ³]	Balling
1	180	800	225	32.1	No
2	200	1000	200	28.6	No
3	200	1200	167	23.8	Yes
4	240	800	300	42.9	No
5	250	1000	250	35.7	No
6	250	1200	208	29.8	Yes
7	300	1000	300	42.9	No

Here, reference pressure p_0 and isotropic exponent γ are chosen to limit density variations to 1% of the reference density ρ_0 .

The set of equations is evolved in time using an explicit Velocity-Verlet time integration scheme [52]. The maximum allowed time-step size follows from the most restrictive stability condition out of the viscous, surface tension, recoil pressure, thermal conduction, and CFL criterion [44,46,53,54].

2.1. Numerical setup

We study single melt track balling of Inconel 718 material at different process conditions. The numerical results are validated with experimental data of Scime and Beuth [38]. In their work, ex-situ recordings of single melt tracks are presented for 36 parameter sets. At each power and scan speed combination 10 single tracks are generated with an EOS M290 machine. The investigated process map covers both stable parameters for constant melt tracks, as well as unstable parameter sets showing defects like balling.

Scime and Beuth use the nickel alloy Inconel 718. The base plate is an Inconel 718 plate from McMaster-Carr (P/N: 1099N8) and the powder material is Inconel 718 from EOS with a composition of Ni 50–55 wt%, Cr 17.0–21.0 wt%, Nb 4.75–5.5 wt%, Mo 2.8–3.3 wt%, Ti 0.65–1.15 wt%. For an accurate representation of the material, temperature dependent material properties are incorporated into the numerical model. The corresponding material models are extracted from [43] and listed in A.3.

In Scime and Beuth [38] a powder layer thickness of $70 \pm 20 \mu\text{m}$ is chosen to achieve a target nominal $40 \mu\text{m}$ fused layer build-up. Thus, a consolidation factor $(1 - \rho_{\text{powder}}/\rho_{\text{fused}})$ of 40% is assumed. No further specifications about the powder size distribution are made. To mimic a comparable powder bed setup, we apply the drop and roll algorithm from Zhou et al. [55]. The initial powder-size distribution of particles to be dropped follows a normal distribution defined by a mean diameter

of $30 \mu\text{m}$, a standard deviation of $10 \mu\text{m}$, and a minimum and maximum particle diameter of $20 \mu\text{m}$ and $50 \mu\text{m}$, respectively. Particles extending beyond a layer height of $70 \mu\text{m}$ are discarded. The resulting powder bed is defined by a packing density of 55%, slightly below the assumed 60% of the experiments.

The same computational setup is used in all simulations. The dimensions of the physical domain are set to $3200 \mu\text{m} \times 300 \mu\text{m} \times 370 \mu\text{m}$ (length, width, height). It is enclosed by three layers of boundary particles. With a uniform spatial discretization of $\Delta x = 3.5 \mu\text{m}$, the total number of SPH particles amounts to 9395040. The laser spot is resolved by about 33 SPH particles across its diameter, and the smallest powder particle is resolved by at least 5 SPH particles across its diameter. In Fig. 1 the simulation setup is shown. At initialization, the domain consists of a $3200 \mu\text{m} \times 300 \mu\text{m} \times 200 \mu\text{m}$ substrate plate, covered by a powder layer of thickness $l_p = 70 \mu\text{m}$. The remainder of the domain is filled with argon, shown in light blue in Fig. 1. In correspondence with the experiments, the domain is initialized at 80°C to resemble preheating. At all domain boundaries, isothermal no-slip boundary conditions are enforced. The laser is initially placed at position $[0, 0, 0]$, as indicated in Fig. 1. Throughout the simulation the laser moves at a constant scan speed u . The simulation is stopped once the laser has travelled $3000 \mu\text{m}$. The laser spot diameter is set in accordance with the experiments to $100 \mu\text{m}$. All investigated process parameter sets of laser power and laser scan speed are listed together with their corresponding linear energy densities $LED = P/u$ and volumetric energy densities $VED = P/(u d_s l_p)$ in Table 1. The quintic spline kernel is applied with a smoothing length equal to the particle spacing and cut-off radius $r = 3.4x$. For a reference length of $21 \mu\text{m}$, reference velocity of 10 m/s and a numerical sound speed of 100 m/s the physical time step size is 7.95 ns .

To allow for high numerical resolutions at moderate computation times, the algorithm utilizes the scaleable OpenFPM C++ framework [56]. The applied domain decomposition, communication, data structures, memory management and cell lists ensure an execution of the comprehensive set of numerical models at reasonable computation times. All simulations were carried out on NVIDIA GeForce RTX 2080 Ti GPUs, and the computational time for $2500 \mu\text{s}$ physical time ranged from 291 to 360 h depending on the process parameter set.

3. Results and discussion

Since melt pool dimensions are of particular importance for the melt breakup process, we first validate them by comparison with experimental data. Next, the ability of the model to predict balling and non-balling parameter sets is assessed. Finally, the melt pool breakup and balling process is investigated to clarify detailed defect mechanisms.

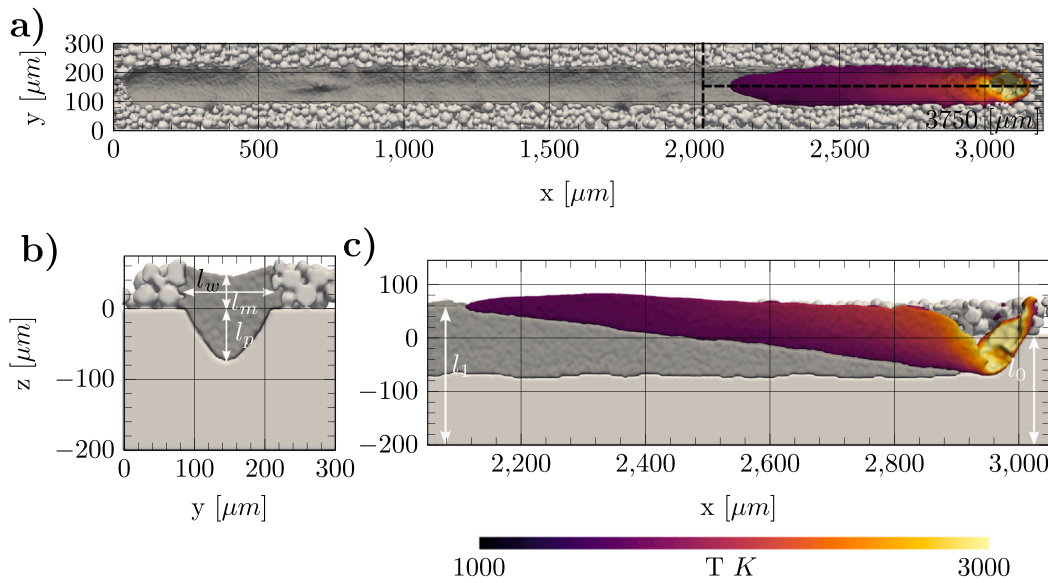


Fig. 2. Single melt track at process conditions #1 (a) in top view, (b) solidified material in transverse cross section view and (c) melt pool in longitudinal cross section view.

3.1. Validation of melt-track dimensions

In order to validate the numerical model, we first compare the accuracy of the simulated single melt tracks to experimental data. The process map from Scime and Beuth [38] suggests a smooth melt pool without balling at the process conditions for cases #1, #2, #4, #5 and #7 in Table 1. Top view, solidified track cross section, and longitudinal melt-pool profile of case #1 are shown as example of a stable single melt track in Fig. 2. At these process conditions a continuous melt pool without significant variance in width develops and is maintained along the scan path. The color coded temperature field, Fig. 2, indicates high temperature within the radiation zone followed by a rapid cooling towards the trailing region of the melt-pool. The longitudinal cross-section view reveals a vapor cavity at the laser irradiation zone followed by a smooth melt-pool topology. The powder layer is fully melted and a sufficient penetration depth into the substrate ensures fusion between the consecutive layers. In the vertical cross-section view the maximum track width l_w , penetration depth into the substrate l_p and consolidated material layer build-up l_m are sketched. The respective total track height follows from the sum of build-up and penetration depth as $l_h = l_p + l_m$. Layer build-up is achieved as the new substrate height l_1 exceeds the initial substrate height l_0 , as shown in Fig. 2 (c).

In Table 2, single-melt-track width and depth results are summarized. For cases #1, #2, #4, #5 and #7 the total track height is stated while for the balling cases #3 and #6 the penetration depth is given. We compare the results with lines of constant melt pool geometry from Scime and Beuth [38]. With a melt track width of 148 μm for case #4, 155 μm for case #7, and track depths of 115 μm and 115 μm for cases #1 and #2, the experimental data are met within an relative deviation of 15 %. Single-track widths and depths of all simulations reside between their corresponding lines of constant melt pool dimension, see Fig. 5 in [38]. Hence, the results indicate the ability of the numerical model to accurately predict the single melt track proportions.

3.2. Balling validation

Next, the occurrence of balling at different process parameter settings is compared with experimental data. In [38], the probabilities of balling at various points in the power-scan-speed map are presented. They are identified from the number of examined cross-sections that exhibit a balling morphology at the specific process parameters. Based

Table 2

Single melt track dimensions.

Serial number	Width l_w [μm]	Depth [μm]	Width [μm] [38]	Depth [μm] [38]
1	129	115		100 ± 7
2	119	115		100 ± 9
3	114	52		
4	148	154	150 ± 7.5	
5	136	145		
6	129	74		
7	155	167	150 ± 7.5	

on these findings, no balling is expected for cases #2 and #5, while balling is likely to occur in cases #3 and #6. Hence, we choose these four parameter sets to assess the numerical prediction of melt pool breakup and balling. Fig. 3 presents the simulation results in the longitudinal cut through the domain center. Case #3 is shown at the time step where the laser is turned off while cases #2, #5 and #6 are shown at the last time steps before the melt pool separation. In all four cases we find melt aggregation and a disturbed track topology in the region of initial melt pool development. At the onset of a single melt track the pool is more likely to accumulate, as shown in [1,25], even when there is no powder bed present [36]. We believe this enhanced accumulation facilitates balling. Therefore, we disregard initial humps and focus the validation on the subsequent melt track. The topologies of cases #2 and #5 remain homogeneous for the most part and only a marginal amount of melt separates in case #2, see Fig. 3 (a) and (b). In contrast, large melt agglomerates separate from the main pool in cases #3 and #6, see Fig. 3 (c) and (d). The developing balls extend over the powder layer height. Thus, the increased likelihood of balling for cases #3 and #6 as found in the experiments is confirmed by the simulations. Besides the indication of balling and non-balling conditions, the results reveal an increased tendency of melt separation, inhomogeneous track topology and disturbed melt pool at higher laser scan speeds and a lower laser power.

3.3. Balling mechanism

The simulations allow for a precise determination of melt pool ratios and to quantify forces relevant for the cause and development of balling. In Fig. 4 and Fig. 5 the melt pool evolution for cases #3 and #6 are shown in top view and as centered cut section, respectively. We distinguish four regions of the melt pool: (I) depression,

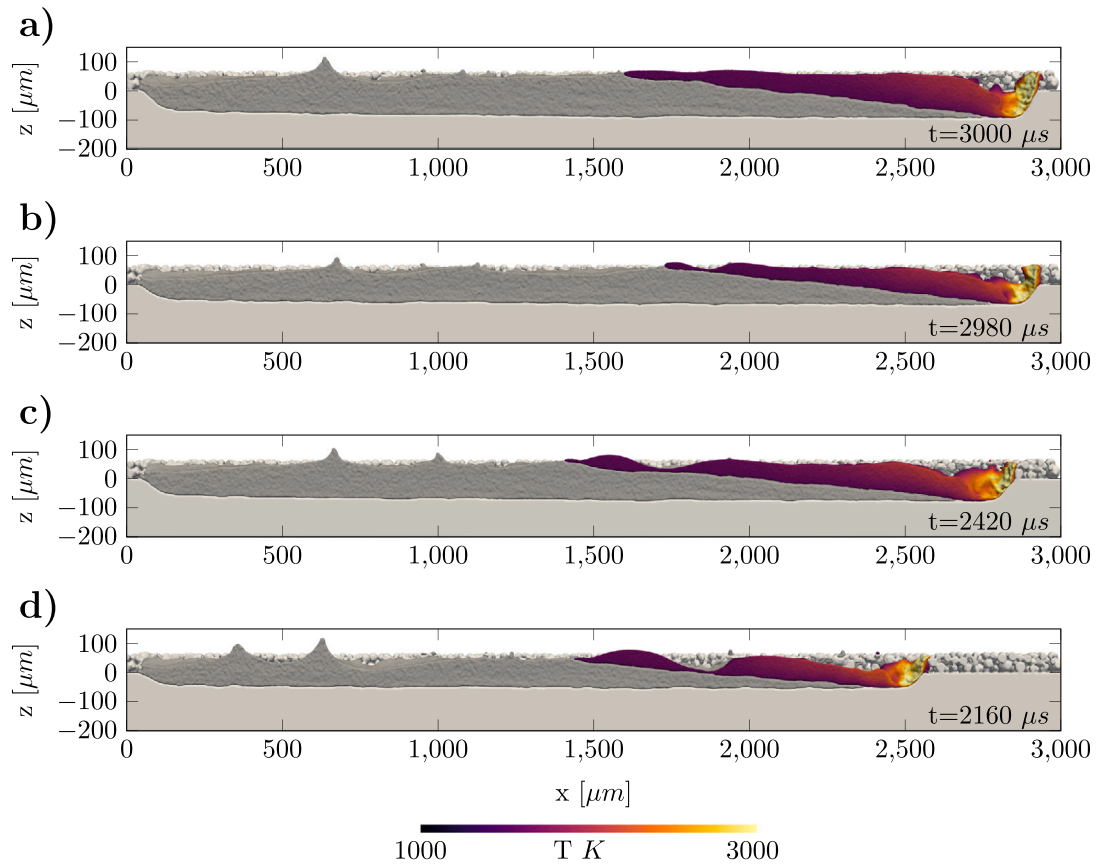


Fig. 3. Longitudinal cross section view of single melt tracks at the last time step before pinch-off for process parameter set (a) #3, (b) #2, (c) #5 and (d) #6.

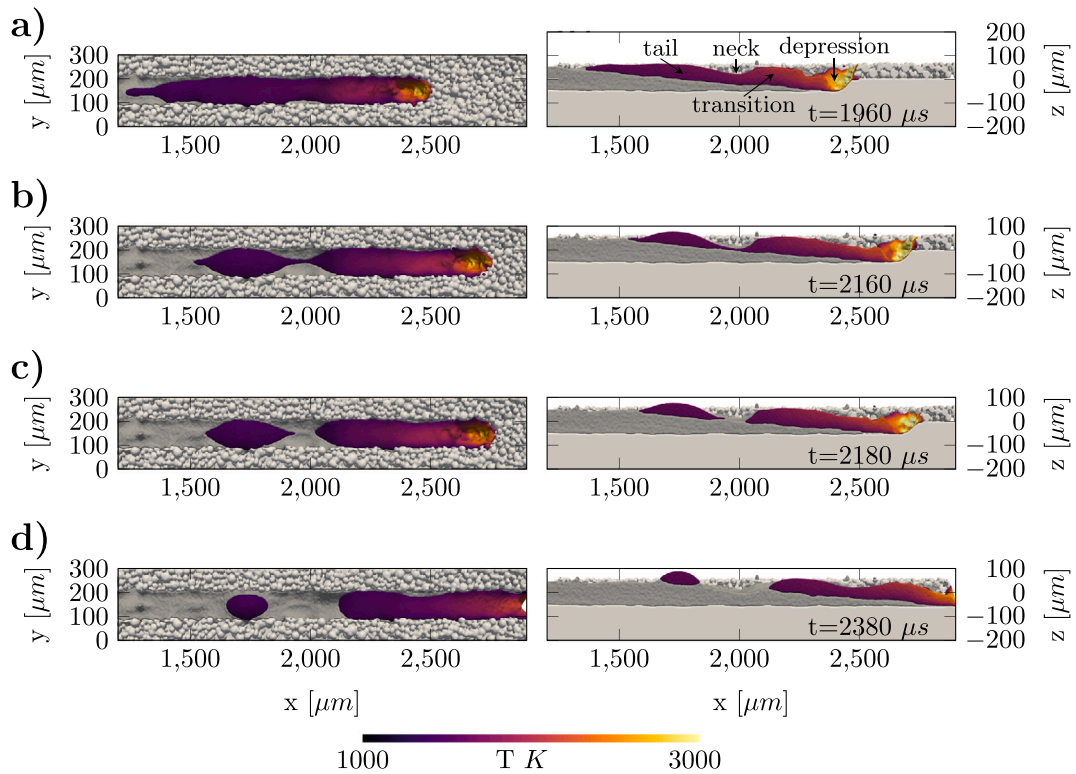


Fig. 4. Top view and longitudinal cross section view of balling process for simulation case #3 at (a) $t = 1960 \mu\text{s}$, (b) $t = 2160 \mu\text{s}$, (c) $t = 2180 \mu\text{s}$ and (d) $t = 2380 \mu\text{s}$.

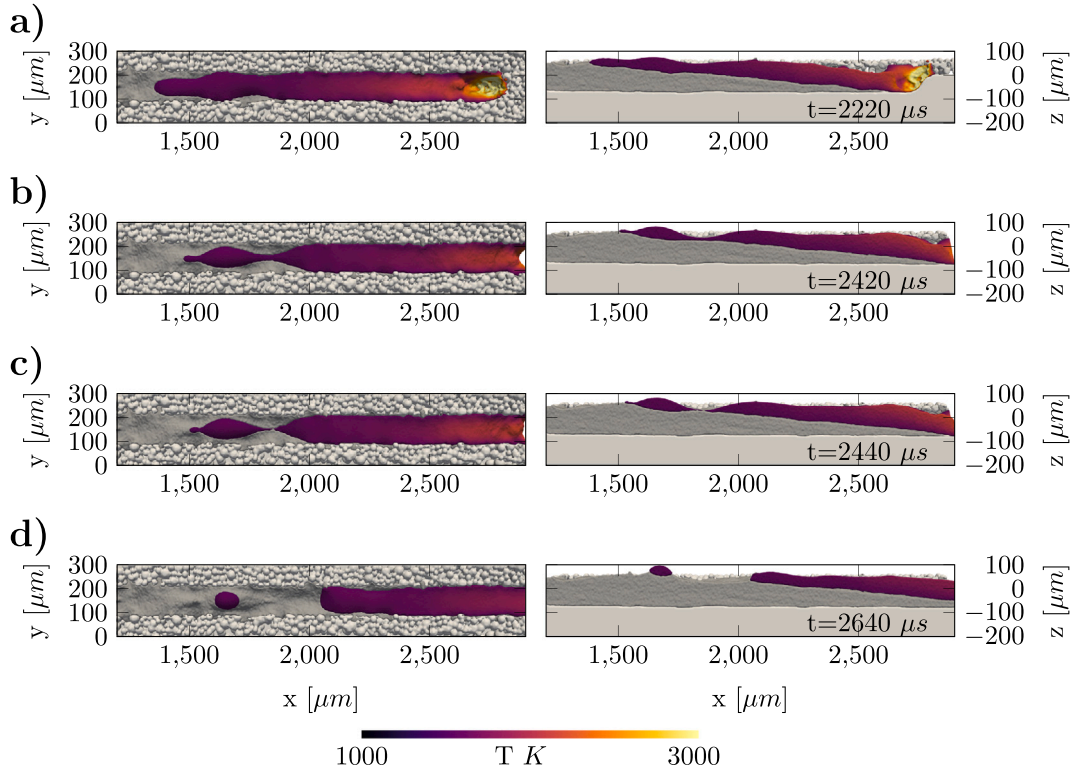


Fig. 5. Top view and longitudinal cross section view of balling process for simulation case #6 at (a) $t = 2220 \mu s$, (b) $t = 2420 \mu s$, (c) $t = 2440 \mu s$ and (d) $t = 2640 \mu s$.

(II) transition, (III) neck, and (IV) tail end. In direct proximity of the laser beam strong recoil pressure and Marangoni forces induce a depression of the melt pool. Further downstream the transition region follows, characterized by decay of the high temperature gradients, the absence of prominent melt extrusion and a constant melt pool width. The section with contracting melt pool post laser-exposure is called the neck-region. The following tail end region eventually forms the isolated ball. Both the depression and transition region appear quasi-steady while the neck and tail region vary periodically to form the emerging balls. The depression constitutes the deepest penetration into the substrate. It is followed by a gradual decrease of melt pool depth in the transition region. The contraction of melt at the tail end leads to a hemispherical shape which exceeds the powder layer height. In contrast, layer build-up in the neck region is marginal. As a result, the solidified track topology consists of a ball followed by a cavity. Note, while the melt pool size of case #6 exceeds the one of case #3 the ball size is smaller in case #6. The ball size depends on the onset of necking. In case #3 necking starts further upstream, leading to a larger tail region and eventually a larger ball. Consequently, the cavity following the ball is more pronounced in case #3 as well.

Melt pool breakup shown in Fig. 4 and Fig. 5 exhibits characteristics of Plateau-Rayleigh instability. A common parameter to evaluate this instability in melt pools is the length to width ratio [7,21]. The length to width threshold ratios found in literature range between 2 and π [13]. A clear indication as to which analytical model is most precise still is missing and their overall applicability is questioned [31]. On the one hand, for process conditions of low melt penetration numerical studies from literature show melt breakup prior to the predicted ratios [10]. On the other hand, the stabilizing effect of melt penetration into the substrate is discarded completely in the analytical models suggesting an underestimation of the breakup ratios [7]. The resulting length to width ratios for cases #3 and #6 are 11.3 and 11.4, respectively. While these ratios agree with the available models they cannot confirm their accuracy as they are well beyond the model thresholds. Large length to width ratios for cases #2 and #5 indicate stable

melt tracks beyond the analytical model predictions. This discrepancy most likely originates from the stabilization due to a considerable melt penetration which is discarded in the analytical models. Nonetheless, the present simulation results confirm the trend of long, shallow and narrow melt pools to be less stable.

Now, we take a closer look at the influence of Marangoni and capillary forces on the necking and breakup behavior. In this work, a linear surface-tension dependence on temperature is incorporated, see σ in A.3. Its impact on the melt pool is twofold. First, the capillary forces normal to the interfaces reduce with temperature. And more importantly, tangential Marangoni forces occur as a result of non-zero surface tension gradients along the interface. In Fig. 6 (b) and (c), case #3 is shown shortly after the onset of necking at time 1960 μs , as well as shortly before the pinch-off at time 2160 μs . The melt pool is color coded with the temperature. The lines ① - ④ and ①' - ④' correspond to the transverse surface tension plots in Fig. 6 (a). In Fig. 6 (d) the surface tension along lines (A) and (B) is shown. The highest heat content and surface-tension gradients are found in the depression region. Downstream, high temperatures at the melt pool center lead to a low surface-tension that rapidly increases towards the cooler melt pool edges. In the transition region the surface tension at the melt pool sides remains constant while it increases along the center. As the Marangoni force results from the surface-tension gradient, their intensity can be estimated from the slope of the surface tension plots. It is concluded that a strong Marangoni force only is present in the depression region and quickly decreases in the transition region. Since surface-tension gradients decay over time, the Marangoni force attenuates as well. Shortly before ball pinch-off, only a weak Marangoni force acts in the neck region. Consequently, the Marangoni force, which promotes melt spreading, decays as necking progresses.

To investigate the impact of capillary forces, we study the evolution of pressure in the neck region over time. Fig. 7 (a) shows the pressure contour of case #3 at times 1960, 2080 and 2140 μs in a cut through the melt pool center. Color coding corresponds to the normalized pressure field. The white x-marked line through the neck region indicates the

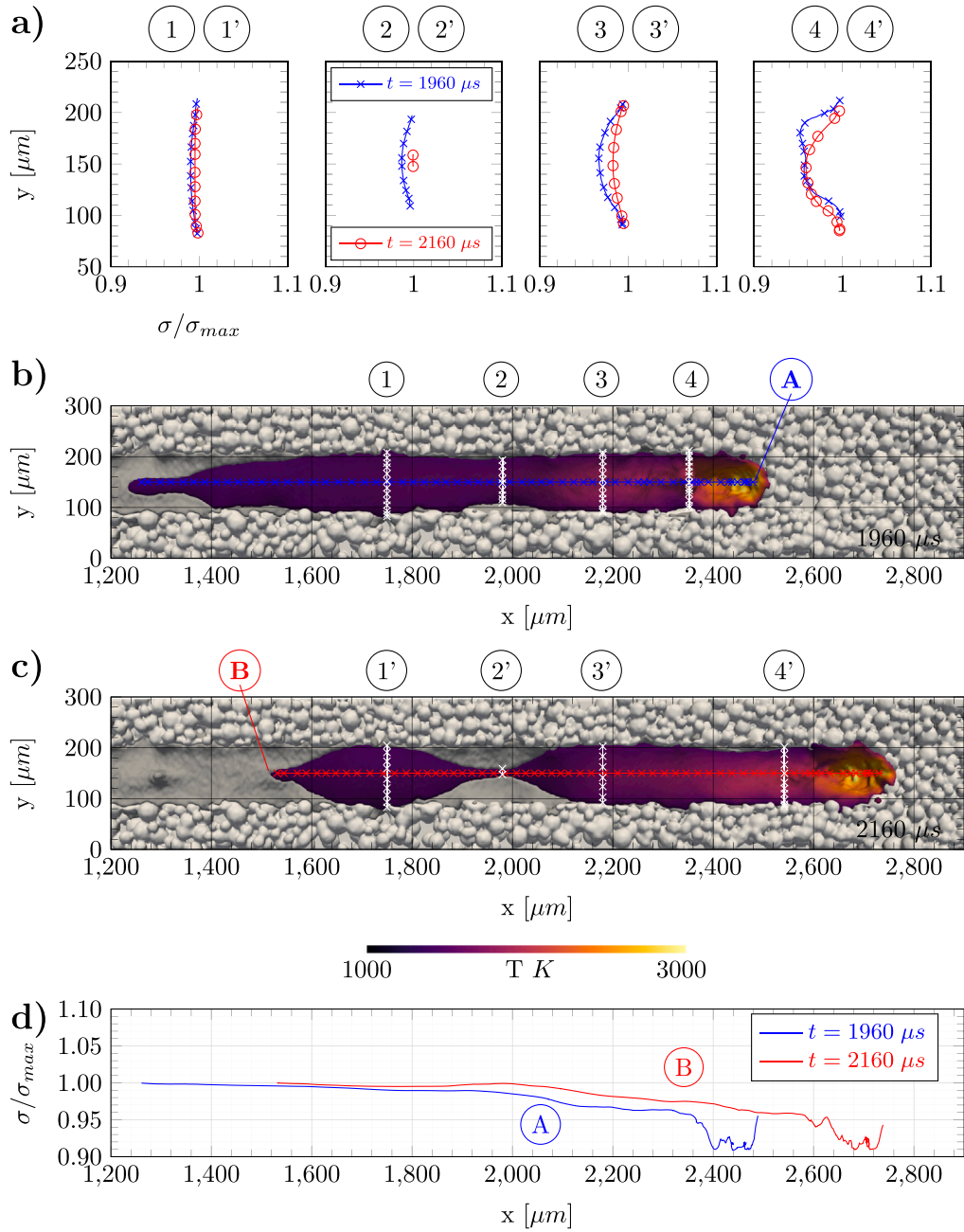


Fig. 6. Surface tension distribution for simulation #3 at time 1960 μs and 2160 μs (a) transverse at locations $x = 1750 \mu\text{m}$, $x = 1980 \mu\text{m}$, $x = 2180 \mu\text{m}$ and $x = 2352 \mu\text{m}$, (b) temperature contour plot shortly after necking onset, (c) temperature contour plot shortly before pinch-off and (d) longitudinal along the melt pool surface.

path of the pressure plot in Fig. 7 (b). As the neck contracts, pressure increases. This is a direct response to a stronger capillary force. As a result, strong pressure gradients arise towards the transition and tail end region. These pressure gradients in turn induce a convective flux outward of the neck region. To quantify the impact of melt convection on the neck contraction and balling, the average specific momentum $I_{sp} = \sum(m_i v_i) / \sum(m_i)$ is plotted over time in Fig. 7 (c), with particle mass m_i and particle velocity v_i . I_{sp} is evaluated in the downstream and upstream neck regions $x = 1750 - 1980 \mu\text{m}$ and $x = 1980 - 2180 \mu\text{m}$, respectively. Shortly before breakup, melt convects towards the neck region. Then, as the pressure in the neck rises, momentum changes direction, and following breakup melt convects outwards from the neck region. Capillary force leads to a pressure increase in the neck that induces melt convection, which induces further necking and ultimately to melt pool breakup.

The melt pool breakup is followed by a melt contraction towards a ball. One common explanation of the balling process is the competing mechanisms of solidification and spreading [8,33,34]. From the assumption of no impact speed of the melted material it is concluded that melt flow is mainly driven by capillary forces and inertia. A comparison of the time scale of super-heat removal to the spreading time is then used to predict the presence or absence of balling. Faster solidification than spreading timescales are claimed to lead to a premature stop of spreading and thus induce balling. In accordance to [33], the solidification time in the present study is approximated by $\tau_{sol} = 2(w^2/3\alpha) \ln(T_0 - T_\infty) / (T_s - T_\infty)$. For cases #3 and #6 follow solidification times of 51 μs and 42 μs . Here, melt track widths of $w = 110 \mu\text{m}$ and $w = 127 \mu\text{m}$, thermal diffusivities of $\alpha = 6.78e^{-6} \text{ m}^2/\text{s}$ and $\alpha = 7.43e^{-6} \text{ m}^2/\text{s}$, super heat temperatures $T_0 = 1579 \text{ K}$ and $T_0 = 1563 \text{ K}$, an ambient temperature of $T_\infty = 353 \text{ K}$ and solidus temperature of

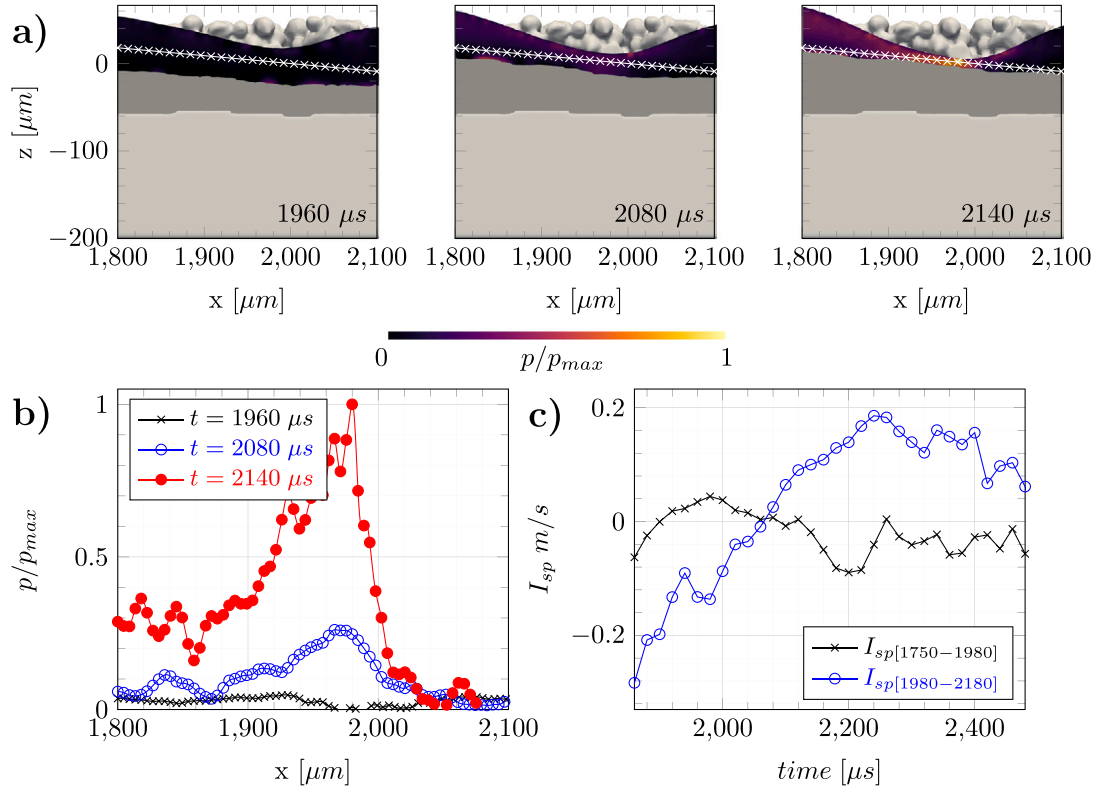


Fig. 7. Longitudinal center cross section in neck region between $x = 1800 \mu m$ and $x = 2100 \mu m$ (a) pressure contour plot at time $t = 1960 \mu s$, $t = 2080 \mu s$ and $t = 2140 \mu s$, (b) normalized pressure distribution through neck and (c) specific momentum in upstream and downstream neck region over time.

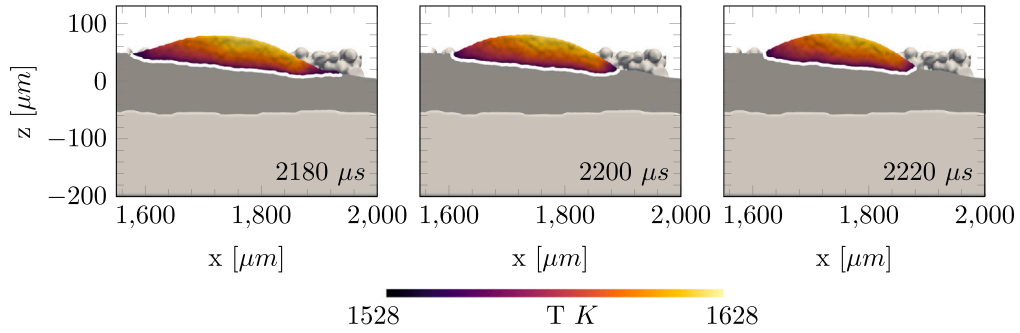


Fig. 8. Balling process of case #3 at time $t = 2180 \mu s$, $t = 2200 \mu s$ and $t = 2220 \mu s$. Solidification front in white, solidified material in dark gray, substrate in light gray, and melt pool temperature color coded. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

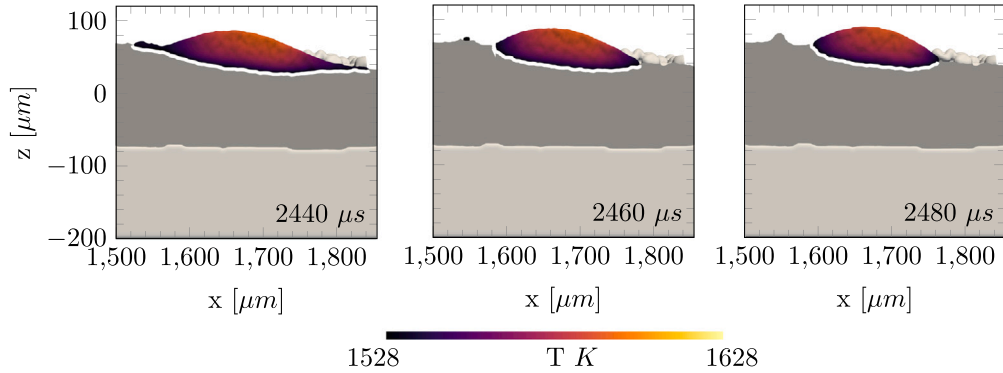


Fig. 9. Balling process of case #6 at time $t = 2440 \mu s$, $t = 2460 \mu s$ and $t = 2480 \mu s$. Solidification front in white, solidified material in dark gray, substrate in light gray, and melt pool temperature color coded. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table A.3
Material properties Inconel 718 [43].

Notation	Property		Value	Units
T_s	Solidus temperature	–	1528	K
T_l	Liquidus temperature	–	1610	K
T_{ev}	Boiling temperature	–	3104	K
ρ_0	Density	–	7733	kg/m ³
ν	Dynamic viscosity	$T_s \leq T$	$0.179 e^{-2} \exp(50.2/(RT))$	Pa s
σ	Surface tension coefficient	$T_s \leq T$	$2.1049 e^{-3} - 0.11 e^{-3} K^{-1} T$	N/m
λ	Thermal conductivity	$T \leq T_s$ $T_s \leq T \leq T_l$ $T_l \leq T$	$5.884 + 0.01654 K^{-1} T$ $111.454 - 0.05255 K^{-1} T$ $4.952 + 0.01360 K^{-1} T$	W/mK
c_p	Specific heat capacity	$0 \leq h < 36367.7$ $36367.7 \leq h < 332707.7$ $332707.7 \leq h < 676972$ $676972 \leq h < 968765$ $968765 \leq h < 2131058$	$334.4 h / (h + 97351.8)$ $493.9 h / (h + 161192.3)$ $652 h / (h + 319284)$ $3558 h / (h + 4759615)$ $778 h / (h + 283854)$	J/kg K
ρ_{el}	Electrical resistivity	$T \leq 1123$ K 1123 K $< T \leq T_s$ $T_s < T \leq T_l$ $T_l < T$	$1.105 e^{-6} + 2.617 e^{-10} K^{-1} T$ $1.341 e^{-6} + 5.182 e^{-11} K^{-1} T$ $4.511 e^{-6} + 6.341 e^{-10} K^{-1} T$ $1.260 e^{-6} + 1.317 e^{-10} K^{-1} T$	Ω m
α	Absorptivity	T	$0.365 \sqrt{\rho_{el}/\lambda} - 0.0667 \rho_{el}/\lambda + 0.006 \sqrt{(\rho_{el}/\lambda)^3}$	–
$h_{ev,n}$	Latent heat of evaporation	–	$6.583 e^6$	J/kg
h_{ev}	Molar heat of evaporation	–	$382 e^3$	J/mol
M	Molar mass	–	$57.96 e^{-3}$	kg/mol

$T_s = 1528$ K are applied, respectively. From $\tau^{spread} = (\rho w^3 / \sigma)^{0.5}$, the spreading timescales $74 \mu\text{s}$ and $92 \mu\text{s}$ follow for cases #3 and #6. Here, density $\rho = 7733 \text{ kg/m}^3$ and surface tension coefficients $\sigma = 1.89 \text{ N/m}$ are used in both cases. The applied material variables are the average values determined in the ball at pinch-off. At that time, most of the melt super-heat is already removed and the tail end spreading is already stopped. As the evaluated spreading time exceeds the solidification time, no melt spreading is expected in both cases.

Overall, the simulation results support the theoretical model. It is interesting to note, that in contrast to the model assumption of zero impact speed, the melt has considerable non-zero momentum after pinch-off, see Fig. 7 (c). In Figs. 8 and 9 the tail end region of cases #3 and #6 and their respective solidification fronts are shown shortly after the melt pool breakup. The melt pool is color coded by temperature. In both cases we find a contracting melt pool rather than a flow dominated by spreading forces. In case #3 the melt contracts from time $t = 2180 \mu\text{s}$ to time $t = 2200 \mu\text{s}$ at which point the solidification front is caught up. The same behavior is found in case #6 where melt contraction and solidification define the balling process. We conclude that, when balling occurs following a necking process, the assumption of zero impact speed of the model by [33] no longer is valid. Instead, the balling process is best characterized by an advancing solidification front at the rear of the tail region and melt contraction followed by an advancing solidification at the front of the tail region.

4. Conclusions

Single melt track simulations are conducted at seven scan-speed-laser-power settings to investigate balling behavior of Inconel 718 material. The present smoothed particle hydrodynamics model accurately predicts multi-phase dynamics including strong interface deformations. The numerical model considers all relevant thermal- and fluid-mechanics effects, including evaporation-induced recoil pressure and cooling as well as capillary, Marangoni and wetting interface forces.

The validity of the model to investigate the balling phenomenon is established by comparison with experimental data. As the breakup is very sensitive to the apparent melt pool dimensions, we compare width and depth with [38] and find good agreement. In addition, the model

capability to predict parameter sets for balling and non-balling melt tracks is validated for two power settings. We identify clearly absence and presence of balls as well as rough and smooth melt pool topologies at the respective conditions.

An investigation of the surface-tension distribution in transversal and longitudinal scan direction reveals the connection of Marangoni forces to the melt pool dynamics. Particular important to the balling process is the decay of surface tension gradients in the necking region during the melt contraction. It indicates a related decay of stabilizing melt flow towards the pool edges.

In contrast, capillary forces within the neck increase as necking progresses both due to an increase of curvature and a reduction in temperature. A resulting high capillary pressure induces a convective melt flux away from the neck region, ultimately leading to the pinch-off of the tail region. As a result, material build-up within the neck area is marginal. Analytical estimates for breakup conditions cannot be confirmed by the numerical results as the melt pool shows to be stable well beyond the model thresholds. This deviation can be explained by stabilizing melt pool penetration into the substrate, which is disregarded by the analytical estimates. We find a comparable breakup behavior for both investigated power settings. The ball size shows to depend more on the necking onset rather than the size of the melt pool itself.

The balling process is best characterized by an advancing solidification front at the rear, and an initial melt contraction at the front that is followed by an advancing melt front at a later stage. Thus, from the simulations the common explanation of the competing mechanisms of spreading and solidification cannot be confirmed. Instead, we find that capillary forces in the neck region induce a melt contraction which is stronger than melt spreading.

CRediT authorship contribution statement

C. Zöller: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **N.A. Adams:** Funding acquisition, Project administration, Resources, Writing – review & editing. **S. Adami:** Conceptualization, Formal analysis, Investigation, Methodology, Software, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

See Table A.3.

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